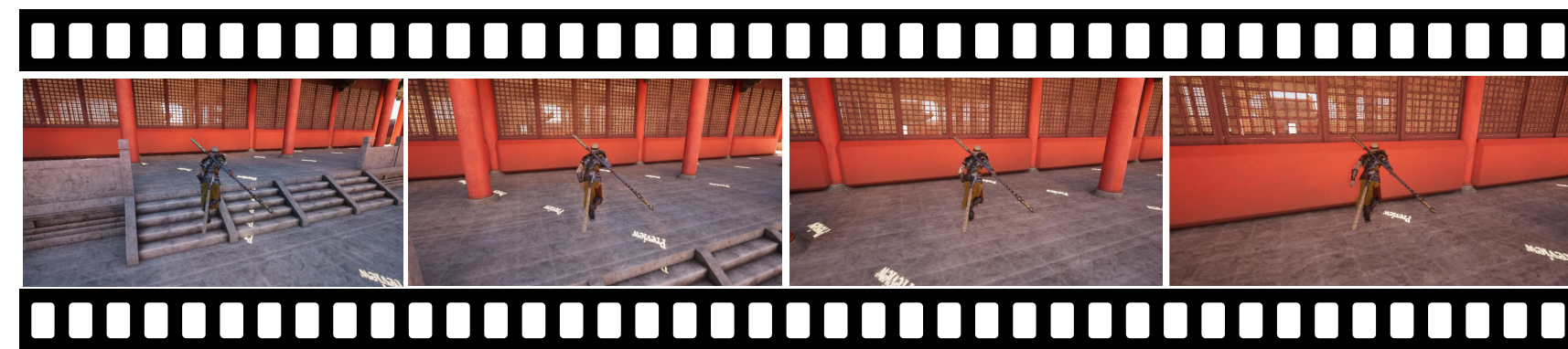
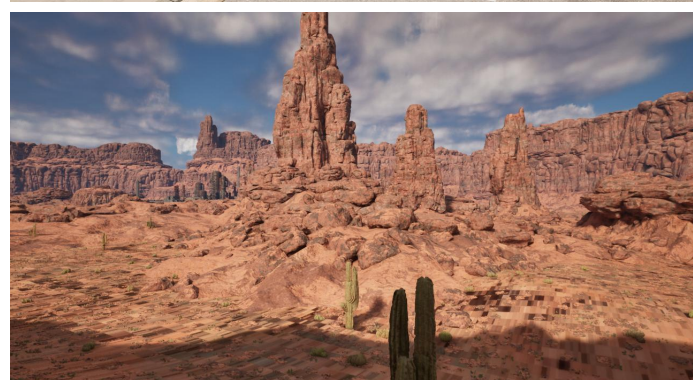
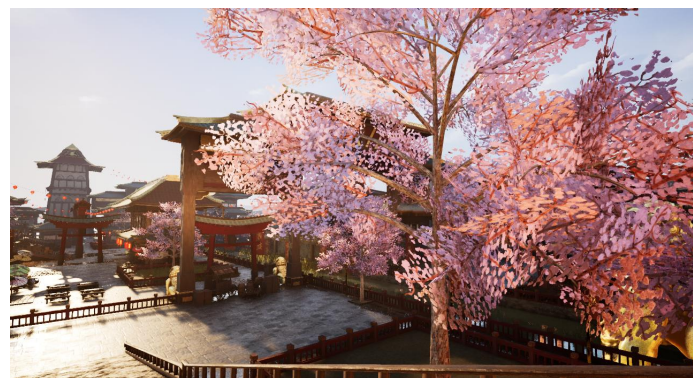




UE Scene Pool

- NavMesh Map
- Map Bounds



1.Exploration & Trajectory Collection

First-Person Free-Camera Engine

Free Camera
Get Location

View Randomizer
Heading +Pitch

Collision Checks
Pose Capture

Quality Gate
Duration + Stuck

Third-Person Free-Camera Engine

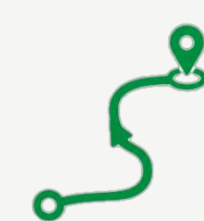
Rigged Actor
NavMesh Path

Smooth Follow
Spring Damper

Camera Move
Sweep & Angle Shift

Visibility Fade
Validation

2.Validation



Validated Trajectories

- Poses
- Actions
- Metadata

3.Offline Rendering



Offline Replay

UE Movie Render Queue



Render Frames
High Quality

4.UE Clips Output



UE Clips

- Frames
- Poses
- Actions
- Metadata

5.Offline Rendering & Runtime Management



Batch Scheduler
Schedule rendering jobs in batches



Ckpt & Resume
Save progress periodically
Resume interrupted sessions



GPU Round-Robin
Distribute jobs across multiple GPUs



Auto Recovery
Detect hangs/timeouts
Restart stuck workers